Dr. Soumyadip Bandyopadhyay

Contact Information

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Professional Summary

Research Scientist and former Assistant Professor with expertise in **Formal Verification, Trustworthy AI, and Control Software Engineering**. My work bridges **formal methods and generative AI** for safe and sustainable software evolution across industrial and cyber-physical systems. I combine academic rigor (35+ peer-reviewed publications) with industrial innovation (IP). Open to opportunities in both **academic research** and **industrial AI/verification labs** that aim to build verifiable, intelligent engineering systems.

Education

Ph.D. in Computer Science and Engineering, IIT Kharagpur, India 2009–2017

- Advisors: Prof. Chittaranjan Mandal, Prof. Dipankar Sarkar
- Thesis: *Path Based Equivalence Checking of Petri Net Representation of Programs for Translation Validation*

B.Tech in Computer Science and Engineering, WBUT, India 2004–2008

- CGPA: 8.43 / 10

Professional Experience

ABB Corporate Research, India — Scientist Jul 2023 – Present
NVIDIA, India — Senior Formal Verification Engineer May 2022 – Jun 2023
BITS Pilani K. K. Birla Goa Campus — Assistant Professor Dec 2018 – May 2022
Hasso Plattner Institute, Potsdam, Germany — Postdoctoral Fellow Aug 2017 – Oct 2018
BITS Pilani K. K. Birla Goa Campus — Assistant Professor Dec 2016 – July 2017
BITS Pilani K. K. Birla Goa Campus — Visiting Faculty Sept 2015 – Dec 2016

Research Vision and Interests

My research aims to develop **trustworthy AI-driven verification frameworks** that integrate formal reasoning into generative engineering workflows. Focus areas include:

- Formal Methods and Software Verification
- AI4Code and AI4Hardware – Generative AI for Software and Hardware Co-Design
- Program Equivalence and Model Checking for Software Evolution
- PLC Verification and Control Software Migration
- Sustainable AI and Hallucination Detection via Formal Analysis

Selected Achievements

- Finalist, Global HackSummit 2024 (ABB)
- ABB Best Employee Award, 2024
- Best Paper Award, ICSOFT 2021
- Bronze Medal, ACM SRC Grand Final 2021 (Mentor)
- Selected among Top 50 Young Researchers, 7th Heidelberg Laureate Forum, 2019
- HPI Postdoctoral Fellowship, 2017
- TCS Innovation Labs Research Fellowship, 2012

Publications Summary

- 3 Journal Papers (Acta Informatica, Parallel Processing Letters)
- 25+ Peer-reviewed Conference Papers (ATVA, FSE, ICSE, MODELS, APSEC etc.)
- 3 Book Chapters (Elsevier)
- 1 Patent

Journals Publications

- **Soumyadip Badyopadhyay**, Dipankar Sarkar, Chittaranjan Mandal, Holger Giese, "Translation Validation of Coloured Petri Net Models of Programs on Integers", **Acta Informatica**
- **Soumyadip Badyopadhyay**, Dipankar Sarkar, Chittaranjan Mandal, "Equivalence checking of Petri net models of programs using static and dynamic cut-points", **Acta Informatica**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar, Kunal Banerjee, Chittaranjan A. Mandal, Krishnam Raju, "A Path Construction Algorithm for Translation Validation using PRES+ Models", **Parallel processing letters**

Conference and workshop Publications

- **Soumyadip Bandyopadhya** and Santonu Sarkar, ”Antarbhukti: Verifying Correctness of PLC Software during System Evolution”, **ATVA 2025**
- **Soumyadip Bandyopadhyay** and Raoul Jetley,”Pn4PLC: Verification of Software Upgrade for PLC Code”, **FSE Companion 2025**
- Md Tauseef Alam, Sorbajit Goswami, Khushi Singh, Raju Halder, Abyayananda Maiti, **Soumyadip Bandyopadhyay**,”SolGen: Secure Smart Contract Code Generation Using Large Language Models Via Masked Prompting”, **ISEC 2025**
- Heiko Koziolek, Virendra Ashiwal, **Soumyadip Bandyopadhyay**, Chandrika K R, ” Automated Control Logic Test Case Generation using Large Language Models”, **ETFA 2024**
- Rakshit Mittal, Dominique Blouin, Anish Bhobe and **Soumyadip Bandyopadhyay**, ”Solving the Instance Model-View Update Problem in AADL”, **MODELS 2022**
- Rakshit Mittal, Dominique Blouin, **Soumyadip Bandyopadhyay**, “PNPEq: Verification of Scheduled Conditional Behavior in Embedded Software”, **APSEC 2021**
- Rakshit Mittal, Rochishnu Banerjee, Dominique Blouin, **Soumyadip Bandyopadhyay**, ”Towards an Approach for Translation Validation of Thread-level Parallelizing Transformations using Colored Petri Nets”, **ICSOFT 2021 (Best Paper)**
- Rakshit Mittal, Rochisnu Banerjee, Santonu Sarkar, **Soumyadip Bandyopadhyay**,” Translation Validation of Loop involving Code Optimizing Transformations using Petri Net based Models of Programs”, **Petri Nets workshop 2020**
- Shivam, Nilanjana Goswami, Veeky Baths, **Soumyadip Bandyopadhyay**, ”AES: Automated Evaluation Systems for Computer Programing Course”, **ICSOFT 2019**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar, Chittaranjan Mandal, ”SamaTulyataOne: A Path Based Equivalence Checker”, **ISEC 2019**
- Santonu Sarkar, Prateek Kandelwal, **Soumyadip Bandyopadhyay**, Holger Giese, ”Analysis of GPGPU Programs for Data-race and Barrier Divergence”, **ICSOFT 2018**
- **Soumyadip Bandyopadhyay**, Santonu Sarkar, Dipankar Sarkar and Chittaranjan Mandal; SamaTulyata, ”An Efficient Path Based Equivalence Checking Tool”, **ATVA 2017**
- **Soumyadip Bandyopadhyay**, Santonu Sarkar and Kunal Banerjee, ”An End-to-End Formal Verifier for Parallel Programs”, **ICSOFT 2017**
- **Soumyadip Bandyopadhyay** and Kunal Banerjee, ”PRESGen: A Fully Automatic Equivalence Checker for Validating Optimizing and Parallelizing Transformations”, **HPDC workshop 17**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar and Chittaranjan Mandal, ”An efficient path based equivalence checking for Petri net based models of programs”, **ISEC-2016**
- **Soumyadip Bandyopadhyay** and Kunal Banerjee, ”Implementing an Efficient Path Based Equivalence Checker for Parallel Programs”, **HPDC workshop 16**
- Kunal Banerjee, **Soumyadip Bandyopadhyay**, and Santonu Sarkar, ”Data-Race Detection: The Missing Piece for an End-to-End Semantic Equivalence Checker for Parallelizing Transformations of Array-Intensive Programs”, **PLDI workshop 2016**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar and Chittaranjan Mandal, ”Validating SPARK: High Level Synthesis compiler”, **ISVLSI-2015**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar and Chittaranjan Mandal, ”A Path-Based Equivalence Checking Method for Petri net based Models of Programs”, **ICSOFT-EA-2015**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar, Chittaranjan A. Mandal, ”An Efficient Equivalence Checking Method for Petri net based Models of Programs”, **ICSE 2015**
- **Soumyadip Bandyopadhyay**, Kunal Banerjee, Dipankar Sarkar, Chittaranjan A. Mandal,” Translation Validation for PRES+ Models of Parallel Behaviours via an FSMD Equivalence Checker”, **VDAT 2012**

Poster Publications

- **Soumyadip Bandyopadhyay**, ”Behavioural verification using Petri net based models of programs”, **POPL-2015 (ACM student research competition)**
- **Soumyadip Bandyopadhyay**, Dipankar Sarkar, Chittaranjan A. Mandal, ”Translation Validation using Path-Based Equivalence Checking of Petri net based Models of Programs”,**WEPL 2015**

Book chapter

- Bedir Tekinerdogan, Rakshit Mittal, Rima Al-Ali, Mauro Iaconod, Eva Navarroe, **Soumyadip Bandyopadhyay**, Ken Vanherpen and Ankica Barisic, ”A feature-based ontology for cyber physical systems”, **Chapter 3, Book Title : Multi-Paradigm Modelling Approaches for**

Cyber-Physical, Elsevier Press. ISBN No. 9780128191064

- Holger Giese , Dominique Blouin , Rima Al-Ali , Hana Mkaoua, **Soumyadip Bandyopadhyay**, Mauro Iacono, Moussa Amrani, Stefan Klikovits and Ferhat Erata "An ontology for multiparadigm modelling", **Chapter 4, Book Title: Multi-Paradigm Modelling Approaches for Cyber-Physical, Elsevier Press. ISBN No. 9780128191064**
- Dominique Blouin, Rima Al-Ali, Holger Giese, Stefan Klikovits, **Soumyadip Bandyopadhyay**, Ankica Barisic and Ferhat Erata, "An integrated ontology for multi-paradigm modelling for cyberphysical systems" , **Chapter 5, Book Title: Multi-Paradigm Modelling Approaches for Cyber-Physical, Elsevier Press, ISBN No. 9780128191064**

Intellectual property (IP)

- **Soumyadip Bandyopadhyay**, Gautam Kumar Pandey, Raoul Jetley, Avijit Mandal, Rahul Singh, Jegan S, "Validating Migrated Code Using Equivalence Checking", Defensive Publication (Accepted May 2025, http://ip.com/IPCOM/000276227)

Industrial R&D Projects

- Formal verification for AI supercomputers and compilers.
- Hallucination detection in LLM-generated PLC control logic.
- Code equivalence for legacy and next-gen industrial systems.
- GenAI-based engineering copilots for control software generation.

Academic Grants and Consultancy Projects

- "APP-based Learning for Python Programming", 6th Sense (15.8L, 2020–2022)
- "Modeling and Verification of Bio-Inspired Systems", DST BIO-CPS (20L, 2021–2022)
- "AES: Automated Evaluation System for Programming Courses", BITS Pilani (2L, 2018–2021)

Tools and Systems Developed

- **Antarbhukti:** Verification tool for PLC software evolution (ATVA 2025)
- **SamaTulyta4PLC:** Verification tool for PLC software migration
- **ABB Co-Pilot:** LLM-driven control logic generation and test synthesis
- **SamaTulyata:** Program equivalence checker based on Petri Nets (ATVA 2017)
- **AutoVal:** Automated evaluation of student programs using equivalence checking

Technical Expertise

Programming: Python, C, C++, Verilog, Shell, LaTeX
Formal Tools: KLEE, CBMC, JasperGold, VC Formal, SAL, Pluto, Par4All, CPN Tools
GenAI Models: Azure OpenAI, LLaMA, Gemini, Claude
Platforms: Linux, Git, Azure ML

Professional Activities

- Reviewer: CAV, DAC, EMSOFT, Acta Informatica, ACM TOSEM
- PC Member: ICSOFT (2018–2023), ISEC, INDICON, VLSI-D, MPM4CPS, BuildSys
- Co-chair: PERR Workshop, FLoC 2022
- IEEE Member, ACM Member, INSTICC Member

Referees

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